

**MATHEMATICS**  
**Introduction to Modern Algebra II**  
**Midterm 3.**

1) Compute the cyclotomic polynomial  $\phi_{15}(x) \in \mathbb{Z}[x]$ .

*Proof.*  $\phi_{15}(x) = (x^{15} - 1)/(\phi_1(x)\phi_3(x)\phi_5(x)) = (x^{15} - 1)/((x^5 - 1)(x^2 + x + 1)) = (x^{10} + x^5 + 1)/(x^2 + x + 1) = x^8 - x^7 + x^5 - x^4 + x^3 - x + 1$ .  $\square$

2) Is the reduction  $\bar{\phi}_{15}(x) \in \mathbb{Z}_2[x]$  of  $\phi_{15}(x)$  modulo 2 irreducible? (Hint: use 1) and the fact that  $F_{16}$  is the splitting field of  $x^{16} - x \in \mathbb{Z}_2[x]$ .)

*Proof.*  $\bar{\phi}_{15}(x)$  divides  $x^{16} - x \in \mathbb{Z}_2[x]$  and hence it splits in  $F_{16}$ . It follows that if  $a$  is any root of  $\bar{\phi}_{15}(x)$ , then  $[\mathbb{Z}_2(a) : \mathbb{Z}_2] \leq [F_{16} : \mathbb{Z}_2] = 4$ . But if  $\bar{\phi}_{15}(x)$  is irreducible, then it is the minimal polynomial of  $a$  and so  $[\mathbb{Z}_2(a) : \mathbb{Z}_2] = 8$  a contradiction. Thus  $\bar{\phi}_{15}(x)$  is reducible in  $\mathbb{Z}_2[x]$ .  $\square$

3) Define the Euler  $\phi$ -function and show that if  $(m, n) = 1$ , then  $\phi(nm) = \phi(m)\phi(n)$ .

*Proof.*  $\phi(n) = \{1 \leq x \leq n - 1 \mid (x, n) = 1\} = |\mathbb{Z}_n^*|$ .  $\phi(nm) = \{1 \leq x \leq nm - 1 \mid (x, nm) = 1\} = \{1 \leq x \leq mn - 1 \mid (x, n) = 1, (x, m) = 1\}$ . If  $(x, n) = 1$ , then  $a = [x]_n$  the residue of  $x$  modulo  $n$  is in  $\mathbb{Z}_n^*$  so there are  $\phi(n)$  possibilities. Similarly if  $(x, m) = 1$ ,  $b = [x]_m \in \mathbb{Z}_m^*$  and there are  $\phi(m)$  possibilities. By the Chinese Remainder Theorem for any fixed  $a \in \mathbb{Z}_n$  and  $b \in \mathbb{Z}_m$  there is a unique  $0 \leq x \leq mn - 1$  such that  $[x]_n = a$  and  $[x]_m = b$ . Thus  $\phi(mn) = |\mathbb{Z}_n^*| \cdot |\mathbb{Z}_m^*| = \phi(n)\phi(m)$ .

**Alternatively** you can argue that by the CRT, we have an isomorphism  $\mathbb{Z}_{nm} \rightarrow \mathbb{Z}_n \times \mathbb{Z}_m$  which induces a bijection  $\mathbb{Z}_{nm}^* \rightarrow \mathbb{Z}_n^* \times \mathbb{Z}_m^*$ , thus  $\phi(mn) = \phi(n)\phi(m)$ .  $\square$

4) Let  $\Theta_m$  and  $\Theta_n$  be primitive roots of 1 such that  $(m, n) = 1$ . Compute the degree  $[\mathbb{Q}(\Theta_m) : \mathbb{Q}]$  and the field  $\mathbb{Q}(\Theta_m, \Theta_n)$ .

*Proof.* The irreducible polynomial of  $\Theta_m$  is  $\phi_m(x)$  which has degree  $\phi(m)$ . Thus  $[\mathbb{Q}(\Theta_m) : \mathbb{Q}] = \phi(m)$ .

We claim that  $\mathbb{Q}(\Theta_m, \Theta_n) = \mathbb{Q}(\Theta_{mn})$ . Since  $\Theta_m = \Theta_{mn}^n$  and  $\Theta_n = \Theta_{mn}^m$ , the inclusion  $\subset$  is clear. Since  $(m, n) = 1$ , there are integers  $a, b$  such that  $an + bm = 1$ . Let  $\Theta = \Theta_n^b \Theta_m^a$  then  $\Theta = e^{2\pi i b/n} e^{2\pi i a/m} = e^{2\pi i (b/n + a/m)}$ . Note that  $b/n + a/m = (bm + an)/mn = 1/mn$  and so  $\Theta$  is a primitive  $mn$ -th root of 1.

**Alternatively**  $\mathbb{Q}(\Theta_m, \Theta_n) = \mathbb{Q}(\Theta_m \Theta_n)$ . In fact  $(\Theta_m \Theta_n)^m = \Theta_n^m$  and since  $\Theta_n$  is a primitive  $n$ -th root of 1 and  $(m, n) = 1$ , then  $\Theta_n^m$  is a primitive  $n$ -th root of 1, thus  $\mathbb{Q}(\Theta_n) = \mathbb{Q}(\Theta_n^m) \subset \mathbb{Q}(\Theta_m \Theta_n)$  and

similarly  $\mathbb{Q}(\Theta_m) \subset \mathbb{Q}(\Theta_m \Theta_n)$ . Thus  $\mathbb{Q}(\Theta_m, \Theta_n) \subset \mathbb{Q}(\Theta_m \Theta_n)$ . The reverse inclusion is trivial and we are done.  $\square$

5) a) Prove that  $\mathbb{Z}_7[i]$  is a field. b) Find a generator for  $\mathbb{Z}_7[i]^*$ . c) How many generators does  $\mathbb{Z}_7[i]^*$  have?

*Proof.*  $x^2 + 1$  has no roots in  $\mathbb{Z}_7$  and hence  $\mathbb{Z}_7[i] \cong \mathbb{Z}_7[x]/(x^2 + 1)$  is a field.  $\mathbb{Z}_7[i]^*$  is a cyclic multiplicative group with 48 elements. Thus it has  $\phi(48) = \phi(2^4)\phi(3) = 8 \cdot 2 = 16$  generators. The element  $2 + i$  is a generator. In fact if it were not a generator, then  $(2 + i)^{16} = 1$  or  $(2 + i)^{24} = 1$ . However,  $(2 + i)^2 = 3 + 4i$ ,  $(2 + i)^4 = 3i$ ,  $(2 + i)^8 = -2$ ,  $(2 + i)^{12} = i$ ,  $(2 + i)^{16} = 4$  and  $(2 + i)^{24} = -1$ .  $\square$